ECE 329 | Fields and Waves I | Summer 19

Kudeki

Exam 1

Wed, June 27, 2019 — 11 AM - 11:55 AM

Name:	Solution
Section:	Main

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is an open book exam but calculators are not allowed. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. Two parts of this question are independent:

a) We have two infinite slabs of space charge of finite widths in the z-direction. Slab 1 occupies the region $-2 < z < -1$ m and has a charge density of $\rho_1 = 4$ C/m^3 while Slab 2 occupies the region $1 < z < 2$ m and has a charge density of $\rho_2 = -4$ C/m^3.

i. (5 pts) What is the electric field vector $\mathbf{E}$ on the z-axis at $z = -3$, -1.5, 0, 1.5, 3 m?

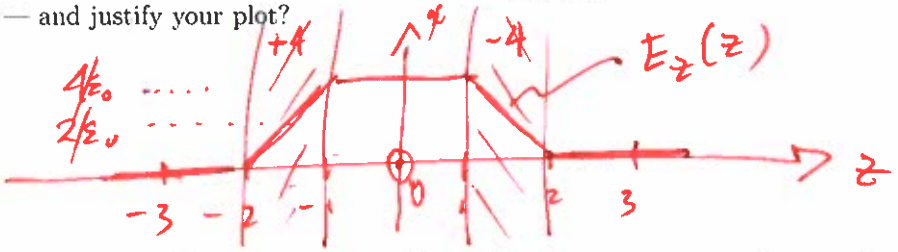
$E_z(-3) = E_z(3) = 0$

$E_z(-1.5) = E_z(+1.5) = 2/\varepsilon_0$ V/m

$E_z(0) = 4/\varepsilon_0$ V/m

} where $\varepsilon_0 \cong \frac{1}{36\pi \times 10^9}$

ii. (5 pts) Plot $E_z(z)$ as a function of z for $-3 < z < 3$ m — consistent with answers to part (i) — and justify your plot?

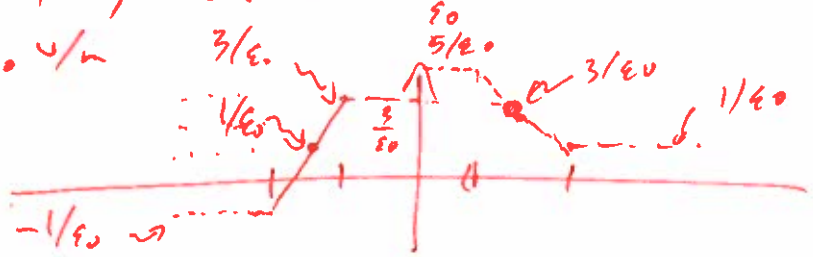


iii. (5 pts) How do the answers to parts (i) and (ii) are changed if a surface charge sheet with 2 C/m^2 charge density is placed on $z = 0$ surface?

$E_z(-3) = -\frac{1}{\varepsilon_0}$ V/m, $E_z(+3) = \frac{1}{\varepsilon_0}$ V/m

$E_z(-1.5) = \frac{1}{\varepsilon_0}$ V/m, $E_z(+1.5) = \frac{3}{\varepsilon_0}$ V/m

$E_z(0) = 4/\varepsilon_0$ V/m

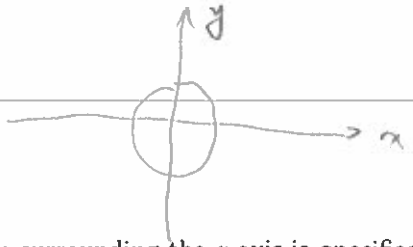


b) (10 pts) Given that $\nabla \times \nabla \times \mathbf{Q} = \nabla(\nabla \cdot \mathbf{Q}) - \nabla^2\mathbf{Q}$. show how and with what conditions an arbitrary vector field $\mathbf{F}$ can be expressed as $\mathbf{F} = \nabla\Phi - \nabla \times \mathbf{R}$ — what are $\mathbf{F}$, Φ, and $\mathbf{R}$ in terms of $\mathbf{Q}$?

$$\nabla^2\bar{Q} = \nabla(\nabla\cdot\bar{Q}) + \nabla\times(\nabla\times\bar{Q})$$

$\bar{F}$ $\quad$ $\bar{\Phi}$ $\quad$ $\bar{R}$

$$\begin{cases} \bar{F} = \nabla^2\bar{Q} \\ \bar{\Phi} = \nabla\cdot\bar{Q} \\ \bar{R} = \nabla\times\bar{Q} \end{cases}$$



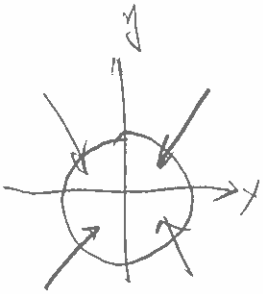
2. Two parts of the following question are independent:

a) Electrostatic potential in the region $r \equiv \sqrt{x^2+y^2} < 2$ m surrounding the z-axis is specified as a 2D function

$$V = \frac{1}{2}(x^2+y^2)\ \text{V}. \quad = \quad \frac{1}{2}x^2 + \frac{1}{2}y^2 .$$

i. (2 pts) What is the location of minimum potential in this region? Explain. $V = 0$

when $x = y = 0 \Rightarrow (x, y, z) = (0, 0, z)$ minimum is reached.
Because otherwise $V > 0$ with any other value of x, y ✓



ii. (3 pts) Based on symmetry arguments, describe the direction of the electric field in this region.

Electric field points from higher to lower potential.
Therefore, $E = -\nabla V$, $\bar{E}$ field points inwards. ✓

iii. (4 pts) Find the vector expression for the electric field $\mathbf{E}(x, y, z)$ in this region? Check that the result is consistent with part (ii).

$$E = -\nabla V = \left(-\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y}, -\frac{\partial V}{\partial z}\right) = \left(-2\cdot\frac{1}{2}x, -\frac{1}{2}\cdot 2\cdot y, 0\right)$$
$$= (-x, -y, 0).$$
$$= -x\,\hat{x} - y\,\hat{y}\ \text{V/m}$$ ✓

iv. (4 pts) Find the charge density $\rho(x, y, z)$ in this region.

$$\nabla\cdot E = \frac{\rho}{\varepsilon_0} \Rightarrow \quad \rho = \varepsilon_0\,\nabla\cdot\bar{E} = \varepsilon_0\cdot\left(\frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) + 0\right)$$
$$= \varepsilon_0\cdot(-1-1) = -2\varepsilon_0\ \text{C/m}^2$$ ✓

v. (4 pts) What is the maximum electric field strength $|\mathbf{E}|$ within the region $r < 4$ m if we also know that $\rho = 0$ for ~~$r > 2$ m~~?

$$|\bar{E}| = \sqrt{\bar{E}\cdot\bar{E}} = \sqrt{E_x^2 + E_y^2} = \sqrt{x^2+y^2} = r \quad \text{when } r < 2$$

$$|\bar{E}| = k\cdot\frac{1}{r} \quad \text{for } r > 2\ , \quad |\bar{E}|_{max} = 2\ \text{V/m}$$ ✓

b) (8 pts) Determine the electrostatic potential $V(1, 1, 1)$ corresponding to the electrostatic field $\mathbf{E} = 2\hat{x} + 4y\hat{y} - 3z^2\hat{z}$ V/m if $V(0, 0, 0) = 0$ V.

$$E\cdot dl = \left(2\hat{x} + 4y\,\hat{y} - 3z^2\,\hat{z}\right)\cdot\left(\hat{x}\,dx + \hat{y}\,dy + \hat{z}\,dz\right) = -dV$$
$$= 2dx + 4y\,dy - 3z^2\,dz = d\left(2x + 2y^2 - z^3\right) = -dV$$
$$V = -2x - 2y^2 + z^3 + C$$

as $V(0,0,0) = 0$, $\quad 0 = 0 + 0 + 0 + C \Rightarrow C = 0.$

$$V = -2x - 2y^2 + z^3$$

$$V(1,1,1) = -2 - 2 + 1 = -3\ \text{V}$$ ✓

3. Two parts of the following question are independent:

a) (15 pts) Consider the following *spherically* symmetric configuration of composite materials in steady-state equilibrium:



i. The region defined by $r \leq 1$ m, where $r = \sqrt{x^2 + y^2 + z^2}$ is the radial distance from the center of the spherical configuration, has $\epsilon = \epsilon_o$, $\sigma = 10^6$ S/m, and holds a net charge of $Q = -4$ C distributed uniformly over its spherical surface.

ii. A perfect dielectric shell with $\epsilon = 3\epsilon_o$ occupies the region $1 < r < 2$ m.

iii. Region $2 \leq r \leq 3$ m has the same material properties as region $r \leq 1$ m and holds zero net charge.

iv. Region $r > 3$ m is occupied by free space. In all four regions we have $\mu = \mu_o$.

Determine, in all four regions, (a) $\mathbf{D}$, (b) $\mathbf{E}$ and (c) $\mathbf{P}$, *and* (d) the surface charge densities, in C/m^2 units, at each of the three material boundaries at $r = 1$, 2, and 3 m. **Hint:** Make use of Gauss's law in integral form, $\oint_S \mathbf{D} \cdot dS = \int_V \rho dV$, with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_o\mathbf{E} + \mathbf{P}$, and a crucial fact about steady-state fields within conducting materials.

$\frac{-4}{4\pi r^2}$

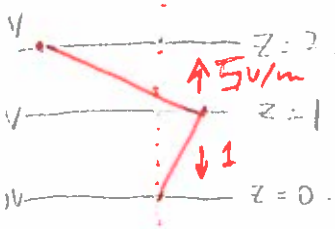
	$r<1$	$r=1$	$1<r<2$	$r=2$	$2<r<3$	$r=3$	$r>3$
D (C/m^2)	0		$-\frac{1}{\pi r^2}\hat{r}$ ✓		0 ✓		$-\frac{1}{\pi r^2}\hat{r}$ ✓
E (V/m)	0		$-\frac{1}{3\varepsilon_0 \pi r^2}\hat{r}$		0		$-\frac{1}{\varepsilon_0 \pi r^2}\hat{r}$
P (C/m^2)	0		$-\frac{2}{3\pi r^2}\hat{r}$		0		0
ρ_s (C/m^2)		$-\frac{1}{\pi}$ ✓		$\frac{1}{4\pi}$ ✓		$-\frac{1}{9\pi}$ ✓	

$P = D - \varepsilon_0 E$ ✓

$-\frac{3}{3\pi r^2} + \varepsilon_0 \cdot \frac{1}{3\varepsilon_0 \pi r^2}$

$-\frac{1}{\pi r^2} + \varepsilon_0 \cdot \frac{1}{\varepsilon_0 \pi r^2}$

b) (10 pts) Conducting plates placed on $z = 0$ and $z = 2$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = 0$ and $V(2) = -4$ V. Also, the potential of the $z = 1$ m surface that supports some surface charge density ρ_{s1} is $V(1) = 1$ V.



i. What is the surface charge densities ρ_{s0} and ρ_{s2} at $z = 0$ and $z = 2$ m, respectively? Justify your answers.

$V(z) = \begin{cases} A_1 z + B_1 & 0 < z < 1 \\ A_2 z + B_2 & 1 < z < 2 \end{cases}$

$(z, V) = (0, 0), (2, -4), (1, 1)$

$0 = B_1 \qquad 1 = A_1$

$-4 = 2A_2 + B_2 \qquad 1 = A_2 + B_2$

$\therefore \begin{matrix} A_1 = 1 & A_2 = -5 \\ B_1 = 0 & B_2 = 6 \end{matrix}$

$\therefore V(z) = \begin{cases} z & 0 < z < 1 \\ -5z + 6 & 1 < z < 2 \end{cases}$

$E(z) = -\nabla V = \begin{cases} -1\,\hat{z} & 0 < z < 1 \\ +5\,\hat{z} & 1 < z < 2 \end{cases}$

$D = E \cdot \varepsilon_0 = \begin{cases} -\varepsilon_0\,\hat{z} & 0 < z < 1 \\ +5\varepsilon_0\,\hat{z} & 1 < z < 2 \end{cases}$

$\rho_{s0} = \hat{n}(D_0^+ - D_0^-)$

$\rho_{s0} = -\varepsilon_0 - 0 = -\varepsilon_0$ C/m^2

$\rho_{s2} = \hat{n}(D_2^+ - D_2^-)$

$\rho_{s2} = 0 - 5\varepsilon_0 = -5\varepsilon_0$ C/m^2 ✓

ii. What is the surface charge density ρ_{s1} at $z = 1$ m? Justify your answer.

$\rho_{s1} = \hat{n}(D_1^+ - D_1^-)$

$\rho_{s1} = +5\varepsilon_0 + \varepsilon_0$

$\rho_{s1} = 6\varepsilon_0$ C/m^2 ✓

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1. Two parts of this question are independent:

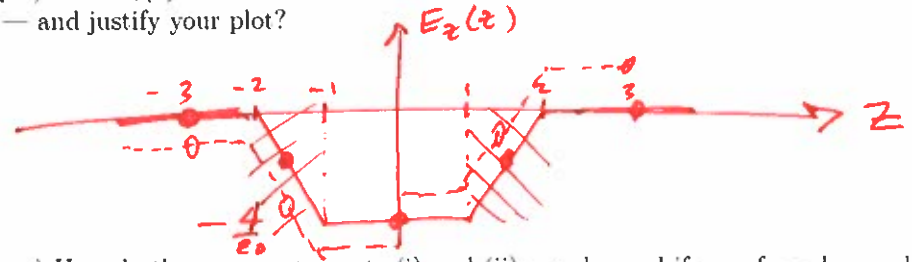
a) We have two infinite slabs of space charge of finite widths in the z-direction. Slab 1 occupies the region $-2 < z < -1$ m and has a charge density of $\rho_1 = -4$ C/m^3 while Slab 2 occupies the region $1 < z < 2$ m and has a charge density of $\rho_2 = 4$ C/m^3.

i. (5 pts) What is the electric field vector $\mathbf{E}$ on the z-axis at $z = -3, -1.5, 0, 1.5, 3$ m?

$0, \frac{-2}{\varepsilon_0}, \frac{-4}{\varepsilon_0}, \frac{-2}{\varepsilon_0}, 0$ @ $z = -3, -1.5, 0, 1.5, 3.0$ m

see

ii. (5 pts) Plot $E_z(z)$ as a function of z for $-3 < z < 3$ m — consistent with answers to part (i) — and justify your plot?



iii. (5 pts) How do the answers to parts (i) and (ii) are changed if a surface charge sheet with 2 C/m^2 charge density is placed on $z = 0$ surface?

With $2\,C/m^2$ charge sheet on $z=0$ surface subtract and add $\frac{1}{\varepsilon_0}$ to above, subtract for $z<0$, add for $z>0$ ...

see dashed plot above

$-\frac{1}{\varepsilon_0}, -\frac{3}{\varepsilon_0}, \frac{4}{\varepsilon_0}, -\frac{1}{\varepsilon_0}, \frac{+1}{\varepsilon_0}$ after the sheet comes.

b) (10 pts) Given that $\nabla \times \nabla \times \mathbf{Q} = \nabla(\nabla \cdot \mathbf{Q}) - \nabla^2 \mathbf{Q}$ and $\mathbf{F} = \nabla\Phi - \nabla \times \mathbf{R}$, what are $\nabla \cdot \mathbf{F}$, $\nabla \times \mathbf{F}$, and $\nabla \cdot \mathbf{R}$ — justify your answers?

with $\bar{F} = \nabla^2 Q$

$\Phi = \nabla \cdot Q$

$R = \nabla \times Q$.

$\Rightarrow \nabla \cdot R = 0$

also $\nabla \cdot F = \nabla \cdot \nabla\Phi + 0 = \nabla^2 \Phi$

$\nabla \times \bar{F} = 0 - \nabla \times \nabla \times \bar{R}$

$\nabla(\nabla \cdot \bar{R}) - \nabla^2 \bar{R}$

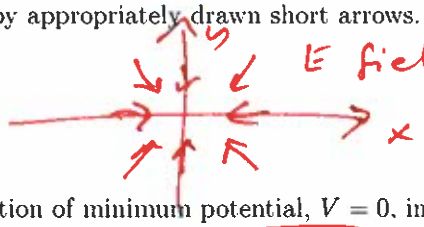
$= \nabla^2 \bar{R}$

2. Two parts of the following question are independent:

a) Electrostatic field in the region $r \equiv \sqrt{x^2+y^2} < 2$ m surrounding the z-axis is specified as

$$\mathbf{E} = -x\hat{x} - y\hat{y}\ \text{V/m}.$$

i. (3 pts) Make a 2D sketch of the vector field $\mathbf{E}$ in the region with the direction and magnitude of $\mathbf{E}$ samples indicated by appropriately drawn short arrows.



E field map.

ii. (2 pts) What is the location of minimum potential, $V = 0$, in this region? Explain.

$V=0$ at $\bar{r}=(0,0,0)$, the origin.

iii. (4 pts) Find the expression for the electrostatic potential $V(x,y,z)$ in this region? Check that the result is consistent with part (ii).

$$\bar{E}\cdot d\bar{l} = -x\,dx - y\,dy = -d\left(\frac{x^2}{2}+\frac{y^2}{2}\right) = -dV$$

$$\therefore V = \frac{1}{2}x^2 + \frac{1}{2}y^2 \ \text{V}.$$

iv. (4 pts) Find the charge density $\rho(x,y,z)$ in this region.

$$\rho_{\varepsilon_0} = \nabla\cdot\bar{E} = \frac{\partial}{\partial x}E_x + \frac{\partial}{\partial y}E_y + \frac{\partial}{\partial z}E_z = -1-1+0 = -2$$

(E_x varies with $-x$, E_y with $-y$, $E_z = 0$)

$$\hookrightarrow \rho = -2\varepsilon_0 \ \text{C/m}^3.$$

v. (4 pts) What is the maximum electric field strength $|\mathbf{E}|$ within the region $r < 4$ m if we also know that $\rho = 0$ for $r > 2$ m?

$$|\bar{E}| = \sqrt{\bar{E}\cdot\bar{E}} = \sqrt{E_x^2 + E_y^2} = \sqrt{x^2+y^2} = r$$

(E_x with $-x$, E_y with $-y$)

$E \propto \frac{1}{r}$ for $r > 2$

$|E|_{max} = r_{max} = 2$ V/m for $0 < r < 2$.

b) (8 pts) Determine the electrostatic potential $V(-1,-1,-1)$ corresponding to the electrostatic field $\mathbf{E} = 2x\hat{x} + 4\hat{y} - 3z^2\hat{z}$ V/m if $V(0,0,0) = 0$ V.

$$\bar{E}\cdot d\bar{l} = 2x\,dx + 4\,dy - 3z^2\,dz = d\left(x^2 + 4y - z^3\right) = -dV$$

$$\therefore V = -x^2 - 4y + z^3$$

$$V(-1,-1,-1) = -(-1)^2 - 4(-1) + (-1)^3 = -1 + 4 - 1 = 2\ \text{V}.$$

3. Two parts of the following question are independent:

a) (15 pts) Consider the following *spherically* symmetric configuration of composite materials in steady-state equilibrium:



i. The region defined by $r \leq 1$ m, where $r = \sqrt{x^2 + y^2 + z^2}$ is the radial distance from the center of the spherical configuration, has $\epsilon = \epsilon_o$, $\sigma = 10^6$ S/m, and holds a net charge of $Q = 2$ C distributed uniformly over its spherical surface.

ii. A perfect dielectric shell with $\epsilon = 4\epsilon_o$ occupies the region $1 < r < 2$ m.

iii. Region $2 \leq r \leq 3$ m has the same material properties as region $r \leq 1$ m and holds a net charge of $Q = 2$ C .

iv. Region $r > 3$ m is occupied by free space. In all four regions we have $\mu = \mu_o$.

Determine, in all four regions, (a) $\mathbf{D}$, (b) $\mathbf{E}$ and (c) $\mathbf{P}$, *and* (d) the surface charge densities, in C/m^2 units, at each of the three material boundaries at $r = 1$, 2, and 3 m. **Hint:** Make use of Gauss's law in integral form, $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \rho dV$, with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_o\mathbf{E} + \mathbf{P}$, and a crucial fact about steady-state fields within conducting materials.

(i) $\bar{D} = \bar{E} = \bar{P} = 0$, $\rho_s|_{r=1} = \frac{2C}{4\pi(1)^2} = \frac{1}{2\pi}\ \frac{C}{m^2}$

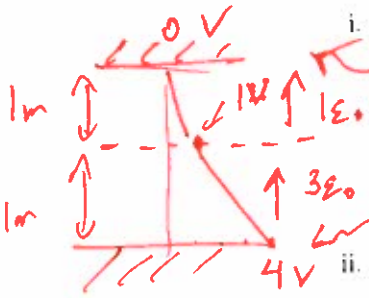
(ii) $\bar{D} = \frac{Q}{4\pi r^2} = \frac{2}{4\pi r^2} = \frac{\hat{r}}{2\pi r^2}\ \frac{C}{m^2}$, $\bar{E} = \frac{\bar{D}}{\epsilon} = \frac{\hat{r}}{8\pi\epsilon_o r^2}\ \frac{V}{m}$

$\bar{P} = \bar{D} - \epsilon_o\bar{E} = \frac{3\hat{r}}{8\pi r^2}\ \frac{C}{m^2}$

(iii) $\bar{D} = \bar{E} = \bar{P} = 0$, $\rho_{s_{in}} = \frac{-2}{4\pi(2)^2} = \frac{-1}{8\pi}\ \frac{C}{m^2}$, $\rho_{s_{out}} = \frac{2}{4\pi(3)^2} = +\frac{1}{18\pi}\ \frac{C}{m^2}$

(iv) $\bar{D} = \frac{2}{4\pi r^2} = \frac{\hat{r}}{2\pi r^2}$, $\bar{E} = \frac{\hat{r}}{2\pi\epsilon_o r^2}$, $\bar{P} = 0$

b) (10 pts) Conducting plates placed on $z = 0$ and $z = 2$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = 4$ and $V(2) = 0$ V. Also, the potential of the $z = 1$ m surface that supports some surface charge density ρ_{s1} is $V(1) = 1$ V.



i. What is the surface charge densities ρ_{s0} and ρ_{s2} at $z = 0$ and $z = 2$ m, respectively? Justify your answers.

$\rho_{s2} = -\epsilon_o\ \frac{C}{m^2}$ //

$\rho_{s0} = 3\epsilon_o\ \frac{C}{m^2}$ //

ii. What is the surface charge density ρ_{s1} at $z = 1$ m? Justify your answer.

$\rho_{s1} = \hat{z}\cdot(1\epsilon_o\hat{z} - 3\epsilon_o\hat{z}) = -2\epsilon_o\ \frac{C}{m^2}$ ///